VICTORIA PARK ACADEMY SHENZHEN | 深圳丽林维育学校

COMPUTER SCIENCE ENRICHMENT PROGRAMME

计算机科学拓展课程计划

Academic Year 2025-2026

2025-2026 学年

Prepared by: Connor Scanlan | Computer Science Lead

Date: September 2025 | 2025 年 9 月

INTRODUCTION | 引言

At Victoria Park Academy Shenzhen, the proposed Computer Science enrichment programme is
designed to ignite curiosity, foster innovation, and prepare students for the digital future. While the academic curriculum builds strong foundations in computational thinking and programming, the enrichment offering goes far beyond the classroom -- providing hands-on experiences, real-world problem solving, and pathways to prestigious universities and careers in technology.

深圳丽林维育学校的计算机科学拓展课程旨在激发好奇心、培养创新精神，并为学生的数字化未来做好准备。虽然我们的学术课程在计算思维和编程方面打下了坚实的基础，但我们的拓展课程远远超出了课堂——提供实践体验、解决真实问题的机会，以及通往知名大学和科技职业的道路。

This proposal
proposal outlines how a dedicated Computer Science teacher will expand and establishes a comprehensive, six-pillar enrichment ecosystem that serves beginners exploring their first lines of code through to elite students competing on the international stage.

本计划阐述了
一名专职计算机科学教师如何将现有优势扩展和增强为一个全面的、由六大支柱组成的拓展课程体系，服务于从初学者到在国际舞台上竞争的精英学生的各个层次。

CORE PROGRAMMES | 核心课程

1. CODING CLUB (BEGINNER & ADVANCED) | 编程俱乐部（初级与进阶）

Schedule: 每周一次，每次 60 分钟 | Weekly sessions, 60 minutes

The Coding Club is the gateway to programming at the school, welcoming students of all ability levels in a supportive, project-driven environment.

编程俱乐部是学校编程学习的门户，在支持性、项目驱动的环境中欢迎

各种能力水平的学生。

BEGINNER TRACK | 初级课程:

• Python fundamentals through interactive games and stories
通过互动游戏和故事学习 Python 基础

• Visual programming transitions to text-based coding
从可视化编程过渡到文本编程

• Project-based learning: build calculators, quizzes, and arcade games
项目式学习：制作计算器、问答游戏和街机游戏

• Collaborative pair-programming exercises
协作结对编程练习

ADVANCED TRACK | 进阶课程:

• Data structures: arrays, linked lists, trees, graphs
数据结构：数组、链表、树、图

• Algorithm design: sorting, searching, recursion, dynamic programming
算法设计：排序、搜索、递归、动态规划

• Competitive programming problem sets and timed challenges
竞赛编程题库和限时挑战

• Introduction to object-oriented and functional programming paradigms
面向对象和函数式编程范式入门

SHOWCASE: End-of-semester Coding Fair where students demonstrate projects to parents, teachers, and the wider VPA community.

展示活动：学期末编程博览会，学生向家长、教师和更广泛的 VPA 社区展示项目。

2. CYBERSECURITY CLUB | 网络安全俱乐部

Schedule: 每周一次，每次 60 分钟 | Weekly sessions, 60 minutes

In an increasingly connected world, cybersecurity literacy is essential. This club demystifies digital security and empowers students to become ethical defenders of the digital realm.

在一个日益互联的世界中，网络安全素养至关重要。本俱乐部揭开数字安全的神秘面纱，培养学生成为数字领域的道德守护者。

CURRICULUM HIGHLIGHTS | 课程亮点:

• Ethical hacking fundamentals and responsible disclosure
道德黑客基础与负责任的漏洞披露

• Password security, encryption, and authentication systems
密码安全、加密和认证系统

• Phishing awareness and social engineering defence

网络钓鱼意识和社会工程防御

• Network security principles and penetration testing basics

网络安全原则和渗透测试基础

HANDS-ON ACTIVITIES | 实践活动:

• Capture-the-Flag (CTF) challenges on platforms like PicoCTF and CyberStart

在 PicoCTF 和 CyberStart 等平台上的夺旗挑战赛

• Security audits of school network infrastructure (with IT permission)

对学校网络基础设施的安全审计（经 IT 部门许可）

• Building secure applications and identifying vulnerabilities

构建安全应用程序和识别漏洞

COMPETITION PATHWAY | 竞赛通道:

• Preparation for CyberFirst Girls and CyberFirst Boys competitions (UK NCSC)

备战 CyberFirst 女子和男子竞赛（英国国家网络安全中心）

• CyberPatriot preparation (US-based cybersecurity competition)

CyberPatriot 竞赛准备（美国网络安全竞赛）

• National cybersecurity Olympiad tracks

国家网络安全奥林匹克赛道

3. HACKATHON TEAM | 黑客马拉松团队

Schedule: 每月一次强化训练，每次 3 小时 | Monthly intensive sessions, 3 hours

The Hackathon Team transforms students into rapid innovators who thrive under pressure. These high-energy sessions simulate real-world tech industry environments.

黑客马拉松团队将学生转变为在压力下蓬勃发展的快速创新者。

这些高强度课程模拟真实的科技行业环境。

PROGRAMME STRUCTURE | 课程结构:

• Real-world problem scenarios from industry partners and social challenges

来自行业合作伙伴和社会挑战的真实问题场景

• Rapid prototyping: from concept to minimum viable product in 3 hours

快速原型设计：3 小时内从概念到最小可行产品

• Team-based collaboration with defined roles (project manager, developer, designer, presenter)

基于团队的协作，明确角色分工（项目经理、开发员、设计师、演讲者）

• Pitch development and stakeholder presentation skills

路演开发及利益相关者演讲技巧

SKILLS DEVELOPED | 培养技能:

• Design thinking and agile development methodologies

设计思维和敏捷开发方法论

• Collaborative coding using Git, GitHub, and version control

使用 Git、GitHub 和版本控制进行协作编程

• Public speaking and persuasive technology pitching

公共演讲和有说服力的技术路演

• Resilience and adaptability in time-constrained environments

在时间受限环境中的韧性和适应性

COMPETITION FOCUS | 竞赛重点:

• Regional and national hackathons (MIT Hackathon, Major League Hacking events)

地区和国家级黑客马拉松（MIT 黑客马拉松、Major League Hacking 赛事）

• Social good hackathons addressing SDGs

解决可持续发展目标的社会公益黑客马拉松

• Inter-school coding competitions

校际编程竞赛

4. DIGITAL CREATORS CLUB | 数字创作者俱乐部

Schedule: 每周一次，每次 60 分钟 | Weekly sessions, 60 minutes

The Digital Creators Club bridges technology and creativity, enabling students to build tangible digital products that serve real communities.

数字创作者俱乐部架起了技术与创造力的桥梁，使学生能够构建服务真实社区的有形数字产品。

PROJECT AREAS | 项目领域:

• Web Development: HTML, CSS, JavaScript, React -- building responsive, accessible websites

网页开发：HTML、CSS、JavaScript、React——构建响应式、无障碍网站

• App Prototyping: Figma, Flutter, React Native for mobile applications

应用原型设计：Figma、Flutter、React Native 移动应用开发

• Digital Design: UI/UX principles, accessibility standards, user research

数字设计：UI/UX 原则、无障碍标准、用户研究

• Real Projects for VPA: school event apps, club websites, digital signage

为 VPA 开发真实项目：学校活动应用、俱乐部网站、数字标识

PORTFOLIO DEVELOPMENT | 作品集开发:

• Each student maintains a GitHub portfolio showcasing their work

每位学生维护展示其作品的 GitHub 作品集

• University application support: personal statements and interview prep

大学申请支持：个人陈述和面试准备

• Industry mentorship connections with Shenzhen tech professionals
与深圳科技专业人士建立行业导师联系
• Annual "Demo Day" presenting projects to parents and tech industry guests
每年"展示日"向家长和科技行业嘉宾展示项目

5. AI & MACHINE LEARNING CLUB | 人工智能与机器学习俱乐部

Schedule: 每两周一次，每次 90 分钟 | Bi-weekly sessions, 90 minutes

Artificial Intelligence is reshaping every industry. This club introduces students to the concepts, tools, and ethical considerations of AI in the context of China's position as a global AI leader.

人工智能正在重塑每个行业。本俱乐部在作为全球人工智能领导者的中国的背景下，向学生介绍人工智能的概念、工具和伦理考量。

CURRICULUM OVERVIEW | 课程概览:

• Introduction to AI: history, types, current capabilities and limitations
AI 导论：历史、类型、当前能力和局限性
• Machine Learning fundamentals: supervised, unsupervised, reinforcement learning
机器学习基础：监督学习、无监督学习、强化学习
• Neural Networks: understanding how deep learning works
神经网络：理解深度学习的工作原理
• AI Ethics: bias, fairness, privacy, and responsible AI development
AI 伦理：偏见、公平、隐私和负责任的 AI 开发

PRACTICAL PROJECTS | 实践项目:

• Image Recognition: building classifiers with TensorFlow and Teachable Machine
图像识别：使用 TensorFlow 和 Teachable Machine 构建分类器
• Chatbots: natural language processing with Python and APIs
聊天机器人：使用 Python 和 API 进行自然语言处理
• Recommendation Systems: building movie/music recommenders
推荐系统：构建电影/音乐推荐器
• Shenzhen Smart City connections: traffic prediction, environmental monitoring
深圳智慧城市关联：交通预测、环境监测

CHINA'S AI STRATEGY CONNECTION | 中国人工智能战略关联:

• Discussion of China's New Generation AI Development Plan
讨论《新一代人工智能发展规划》
• Shenzhen's role as an AI innovation hub
深圳作为 AI 创新中心的角色
• Careers in AI: research, engineering, policy, and ethics
AI 领域的职业：研究、工程、政策和伦理

6. COMPETITIVE PROGRAMMING SQUAD | 竞赛编程小队

Schedule: 每周一次，每次 60 分钟 | Weekly sessions, 60 minutes

The Competitive Programming Squad is VPA's elite training programme for students aspiring to excel in computational thinking competitions. This programme is a significant booster for top university admissions worldwide.

竞赛编程小队是 VPA 为渴望在计算思维竞赛中脱颖而出的学生设立的精英培训项目。该项目是申请全球顶尖大学的重要加分项。

TRAINING COMPONENTS | 培训内容:

- Algorithm mastery: greedy algorithms, divide-and-conquer, graph theory, dynamic programming
  算法精通：贪心算法、分治法、图论、动态规划
- Speed coding drills with increasing time pressure
  在时间压力下进行快速编码训练
- Problem-solving strategy frameworks and pattern recognition
  解题策略框架和模式识别
- Mock competitions under exam conditions
  在考试条件下进行模拟竞赛

COMPETITION TARGETS | 竞赛目标:

- Bebras Computational Thinking Challenge (November)
  Bebras 计算思维挑战赛（11 月）
- UKMT Senior Computing Challenge
  UKMT 高级计算挑战赛
- British Informatics Olympiad (BIO)
  英国信息学奥林匹克竞赛（BIO）
- International Olympiad in Informatics (IOI) preparation pathway
  国际信息学奥林匹克竞赛（IOI）准备通道
- American Computer Science League (ACSL)
  美国计算机科学联赛（ACSL）

UNIVERSITY ADMISSIONS BOOSTER | 大学申请加分项:

- Demonstrates exceptional problem-solving ability to admissions officers
  向招生官展示卓越的解题能力
- Develops mathematical maturity highly valued by Oxbridge and Ivy League
  培养牛津剑桥和常春藤盟校高度重视的数学成熟度
- Builds resilience and intellectual stamina
  培养韧性和智力耐力

• Creates distinctive personal statement content
创造独特的个人陈述内容

QUICK-START IDEAS | 快速启动项目

Beyond the six core programmes, the following activities can be launched quickly to build momentum and engage diverse student interests:
除了六大核心课程外，以下活动可以快速启动，以积累动力并吸引不同兴趣的学生：

7. eSports Strategy & Management | 电竞战略与管理
   -- Team management, event organisation, streaming technology, game analytics
   团队管理、赛事组织、流媒体技术、游戏数据分析
8. Drone Programming | 无人机编程
   -- Python-controlled drones, autonomous flight paths, aerial photography
   Python 控制无人机、自主飞行路径、航拍
9. Arduino & micro:bit Projects | Arduino 与 micro:bit 项目
   -- IoT devices, wearable tech, sensor networks, automated systems
   物联网设备、可穿戴技术、传感器网络、自动化系统
10. Girls in STEM Initiative | STEM 女子计划
   -- Mentorship, guest speakers, confidence-building workshops, all-female coding competitions
   导师指导、嘉宾演讲、信心建设研讨会、全女性编程竞赛
11. Tech Podcast / Blog | 科技播客/博客
   -- Student-run media covering tech news, interviews, tutorials
   学生运营的科技新闻、访谈、教程媒体
12. Digital Art & Animation | 数字艺术与动画
   -- Processing, p5.js, generative art, stop-motion animation
   Processing、p5.js、生成艺术、定格动画
13. 3D Printing & Modelling | 3D 打印与建模
   -- Tinkercad, Blender, Fusion 360 -- design and manufacture physical objects
   Tinkercad、Blender、Fusion 360——设计并制造实体物品
14. Tech Mentorship Programme | 科技导师计划
   -- Older students teach younger students -- leadership and consolidation
   高年级学生教低年级学生——领导力培养和知识巩固
15. Parent Tech Workshops | 家长科技工作坊
   -- Cyber safety for families, AI in education, coding taster sessions
   家庭网络安全、AI 在教育中的应用、编程体验课

10. Community Coding Events | 社区编程活动

-- Weekend coding workshops for local Chinese students, bilingual instruction
为当地中国学生举办周末编程工作坊，提供双语教学

11. App Development for Social Good | 社会公益应用开发

-- Apps addressing mental health, sustainability, accessibility, community needs
解决心理健康、可持续发展、无障碍和社区需求的应用

12. Data Science Projects | 数据科学项目

-- Python data analysis, pandas, visualisation, real datasets from VPA life
Python 数据分析、pandas、可视化、来自 VPA 生活的真实数据集

13. Virtual Reality Exploration | 虚拟现实探索

-- Unity VR development, 360 media, immersive storytelling
Unity VR 开发、360 度媒体、沉浸式叙事

14. Tech Industry Guest Speakers | 科技行业嘉宾演讲

-- Regular visits from Shenzhen tech professionals (Tencent, Huawei, DJI)
定期邀请深圳科技专业人士（腾讯、华为、大疆）到访

15. University Partnership Visits | 大学合作访问

-- Visits to HKUST-Shenzhen, SUSTech, CUHK-Shenzhen CS departments
访问港科大（深圳）、南科大、港中文（深圳）计算机科学系

COMPETITION CALENDAR | 竞赛日历

TERM 1 | 第一学期 (September - January)

September 九月 -- Club recruitment and training begins
俱乐部招募和培训开始

October 十月 -- Hackathon Team: internal practice event
黑客马拉松团队：内部练习赛

November 十一月 -- BEJRAS Computational Thinking Challenge (All students)
Bebras 计算思维挑战赛（全体学生）

-- CyberFirst registration opens
CyberFirst 竞赛报名开始

December 十二月 -- Competitive Programming Squad: mid-year assessment

竞赛编程小队：年中评估

January 一月 -- CyberFirst Girls/Boys competition heats

CyberFirst 女子/男子竞赛初赛

-- First-semester Coding Fair showcase

第一学期编程博览会展示

TERM 2 | 第二学期 (February - July)

February 二月 -- British Informatics Olympiad (BIO) Round 1

英国信息学奥林匹克竞赛（BIO）第一轮

-- Regional hackathon season begins

地区黑客马拉松赛季开始

March 三月 -- UKMT Senior Computing Challenge

UKMT 高级计算挑战赛

April 四月 -- CyberPatriot semi-finals (if qualified)

CyberPatriot 半决赛（如晋级）

-- BIO Round 2 (for qualified students)

BIO 第二轮（针对晋级学生）

May 五月 -- Major League Hacking (MLH) hackathon events

Major League Hacking (MLH) 黑客马拉松赛事

June 六月 -- End-of-year showcase and awards ceremony

年终展示和颁奖典礼

July 七月 -- Summer coding intensive (preparation for next year)

暑期编程强化（为下一年做准备）

ONGOING THROUGHOUT THE YEAR | 全年持续:

• Monthly Competitive Programming practice contests (internal ranking)

每月竞赛编程练习赛（内部排名）

• Quarterly CTF challenges for Cybersecurity Club

网络安全俱乐部季度夺旗挑战赛

• Rolling application development for Digital Creators Club

数字创作者俱乐部滚动应用开发

RESOURCE REQUIREMENTS | 资源需求

SOFTWARE | 软件 (Primarily Free/Open Source | 主要为免费/开源)

• Python 3.x, IDLE, VS Code, PyCharm (Education edition - free)

• Git and GitHub (GitHub Education free for schools)

• Cybersecurity: Kali Linux (free), Wireshark (free), VirtualBox (free)

• AI/ML: TensorFlow (free), Google Colab (free), Jupyter Notebooks (free)

• Web Dev: Node.js (free), Figma (free education), Chrome DevTools (free)

• 3D/VR: Blender (free), Unity (free education), Tinkercad (free)

• Competitive: Codeforces, AtCoder, LeetCode, HackerRank (free tiers)

Total annual software cost: Minimal to zero | 年度软件总成本：极少或为零

HARDWARE | 硬件

Existing Resources | 现有资源:

-- School computer labs and student devices

学校计算机实验室和学生设备

-- Existing equipment (expandable)

现有机器人设备（可扩展）

Requested Additions | 新增需求:

-- 5x Arduino starter kits (RMB 2,000)

-- 5x BBC micro:bit sets (RMB 1,500)

-- 2x Raspberry Pi 4 kits for server/network projects (RMB 1,500)

-- 1x 3D printer for Digital Creators Club (RMB 5,000)

-- Basic drone (programmable, Python SDK) (RMB 3,000)

-- VR headset for exploration (Meta Quest - education pricing) (RMB 3,000)

Total hardware investment: RMB 16,000 (one-time) | 硬件总投资：16,000 元（一次性）

STAFFING | 人员配置

Programme Lead: Connor Scanlan (Computer Science Teacher)

项目负责人：Connor Scanlan（计算机科学教师）

Connor will lead all enrichment sessions, drawing on his subject expertise and experience in competitive programming, web development, and AI education.

Connor 将领导所有拓展课程，运用他在竞赛编程、网页开发和 AI 教育方面的专业知识。

Student Leadership Structure | 学生领导结构:

-- Club Captains (Year 12/13 students): 6 positions, 1 per core club

俱乐部队长（12/13 年级学生）：6 个职位，每个核心俱乐部 1 名

-- Deputy Captains (Year 10/11 students): 6 positions

副队长（10/11 年级学生）：6 个职位

-- Mentors: Advanced students paired with beginners (rotating)

导师：进阶学生与初学者配对（轮换）

This structure develops leadership skills while ensuring programme sustainability and peer-to-peer learning.

该结构培养领导技能，同时确保项目的可持续性和同伴学习。

VISION | 愿景

The vision is to make Victoria Park Academy Shenzhen a regional centre of excellence for Computer Science education -- where every student discovers the power of code, where elite competitors train alongside curious beginners, and where technology serves community and creativity.

我们的愿景是使深圳丽林维育学校成为计算机科学教育的区域卓越中心——让每位学生发现代码的力量，让精英竞赛者与好奇的初学者并肩学习，让技术服务社区和创造力。

With Connor Scanlan's dedicated leadership, VPA's Computer Science enrichment programme will prepare students not just for examinations, but for the digital civilisation they will shape.

在 Connor Scanlan 的专职领导下，VPA 的计算机科学拓展课程将不仅为学生备考做准备，更为他们将塑造的数字文明做准备。

In Shenzhen -- China's Silicon Valley -- there is no better place to learn, create, and lead in technology.

在深圳——中国的硅谷——没有比这更好的地方来学习、创造和引领科技。

VICTORIA PARK ACADEMY SHENZHEN | 深圳丽林维育学校

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